Cache-Resident Cross-Section Reconstruction via Singular Value Decomposition for Monte Carlo Neutron Transport

F. Rodrigues

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Abstract

We present a method for on-the-fly reconstruction of neutron cross-sections using truncated Singular Value Decomposition (SVD) that eliminates the memory-wall bottleneck in continuous-energy Monte Carlo transport. The cross-section matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$, indexed by energy and temperature, is decomposed in log-space as $\log_{10} \mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$, where the pre-multiplied basis $\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k$ fits entirely within the CPU L3 cache. Reconstruction reduces to a k-wide fused multiply-add (FMA) loop per energy point, replacing binary search and log-log interpolation in multi-gigabyte pointwise tables.

Validation against ENDF/B-VII.1 and JEFF-3.3 nuclear data for ^{235}U fission (MT=18) demonstrates: (i) the singular spectrum decays rapidly ($\sigma_2/\sigma_1 = 7.7 \times 10^{-2}$), confirming low effective rank; (ii) a pure-Rust implementation achieves 8–13× speedup over traditional table lookup at 3–5 ns per energy point; (iii) Godiva (HEU-MET-FAST-001) criticality benchmarks with OpenMC show $\Delta k_{\text{eff}} < 10$ pcm at all tested ranks ($k = 2$ through $k = 5$), indistinguishable from the unmodified library within Monte Carlo statistics. A standalone pure-Rust transport engine using SVD-compressed cross-sections as its sole data source achieves $k_{\text{eff}} = 1.00016 \pm 0.00080$ on Godiva—16 pcm from the experimental value—with a complete eigenvalue calculation in 3.4 seconds. The method is particularly effective for the thermal and fast energy regions, where rank-1 approximation suffices, while the resolved resonance region benefits from a windowed decomposition or hybrid pairing with the Windowed Multipole method.

Keywords: Monte Carlo, cross-section compression, singular value decomposition, cache optimization, nuclear data, Rust

1 Introduction

Continuous-energy Monte Carlo neutron transport codes such as OpenMC [1], MCNP, and Serpent rely on pointwise cross-section tables spanning 10^4–10^5 energy points per nuclide and temperature. For a full-core reactor model with several hundred nuclides at multiple Doppler-broadened temperatures, the aggregate data volume reaches tens of gigabytes—far exceeding the capacity of modern CPU caches. Every cross-section lookup thus incurs a cache miss to main memory (100–200 ns latency), creating a *memory wall* that dominates the runtime of particle tracking.

To quantify the scale of the problem: the ENDF/B-VII.1 library contains 444 nuclide files totalling 5.8 GB on disk (the newer ENDF/B-VIII.0 is 13.7 GB due to denser energy grids). Table 1 shows the in-memory footprint for representative nuclides when loaded by OpenMC. A single actinide such as ^{238}U requires 85.5 MB of pointwise data (51 reactions $\times$ 6 temperatures, 5.6 million data points). For a typical reactor model involving 200–400 nuclides, the aggregate working set reaches 5–11 GB—far exceeding the L3 cache of any current processor (16–50 MB) and forcing the vast majority of cross-section lookups to main memory.

Table 1: In-memory pointwise data size per nuclide (ENDF/B-VII.1, 6 temperatures). Data points include energy grid + cross-section value (16 bytes per point).

Nuclide	Role	Reactions	Data points	RAM
^{235}U	Fissile actinide	52 $\times$ 6T	2.6 M	39.7 MB
^{238}U	Fertile actinide	51 $\times$ 6T	5.6 M	85.5 MB
^{239}Pu	Fissile actinide	52 $\times$ 6T	2.5 M	38.1 MB
^{16}O	Moderator/fuel	72 $\times$ 6T	0.3 M	4.3 MB
^{1}H	Moderator	6 $\times$ 6T	0.02 M	0.3 MB
^{90}Zr	Structural	52 $\times$ 6T	0.15 M	2.2 MB
6 nuclides (PWR pin cell)				170 MB
Projected: 50 nuclides				1.4 GB
Projected: 200 nuclides				5.5 GB
Projected: 400 nuclides				11.1 GB

Several strategies have been proposed to mitigate this bottleneck. Machine-learning-based adaptive resampling reduces pointwise grid density while preserving resonance features, achieving $\Delta k_{\text{eff}} < 100$ pcm with 10–50% of the original data [2, 3]. The Windowed Multipole (WMP) method represents cross-sections analytically using multipole parameters, reducing memory by one to two orders of magnitude for the resolved resonance region [4, 5]. Hybrid neural-network approaches combine WMP with learned Doppler broadening kernels, achieving 65% memory reduction [6].

In this work we propose a complementary approach inspired by low-rank compression techniques from the AI/ML community—specifically the KV-cache compression methods used in large language models [7, 8, 9, 10]. We observe that the cross-section matrix, when viewed in log-space, exhibits rapid singular-value decay, and exploit this structure via truncated SVD to build a *cache-resident* reconstruction kernel.

The key contributions are:

1. A pure-Rust implementation that reads OpenMC HDF5 nuclear data, performs SVD decomposition, and reconstructs cross-sections entirely without C/Fortran dependencies.

2. Empirical characterisation of the singular spectrum for ^{235}U, ^{238}U, and ^{239}Pu across two independent nuclear data libraries (ENDF/B-VII.1 and JEFF-3.3).

3. Demonstration that the SVD basis vectors fit in L3 cache, enabling reconstruction via streaming FMA at 3–5 ns per energy point—an order of magnitude faster than table lookup.

4. Godiva criticality benchmark validation showing $\Delta k_{\text{eff}} < 10$ pcm, within Monte Carlo statistical noise.

5. A windowed SVD analysis showing that the thermal ($E < 1$ eV) and fast ($E > 25$ keV) regions are effectively rank-1, while the resolved resonance region requires $k = 5$–6 for sub-percent accuracy.

6. A standalone pure-Rust Monte Carlo transport engine using SVD cross-sections with full physics (anisotropic scattering, tabulated fission spectra, URR probability tables, free-gas thermal model) achieving $k_{\text{eff}} = 1.00016 \pm 0.00080$ on Godiva in 3.4 seconds wall time.

2 Method

2.1 Problem formulation

Let $\sigma(E, T)$ denote the microscopic cross-section (barns) for a given nuclide and reaction, evaluated on a discrete energy grid $\{E_i\}_{i=1}^{N_E}$ at temperatures $\{T_j\}_{j=1}^{N_T}$. We form the matrix

$$\mathbf{A} \in \mathbb{R}^{N_E \times N_T}, \quad A_{ij} = \sigma(E_i, T_j), \tag{1}$$

and work in log-space to normalise the six-order-of-magnitude dynamic range:

$$\tilde{\mathbf{A}} = \log_{10} \mathbf{A}. \tag{2}$$

2.2 SVD decomposition

The thin SVD of $\tilde{\mathbf{A}}$ yields

$$\tilde{\mathbf{A}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \tag{3}$$

where $\mathbf{U} \in \mathbb{R}^{N_E \times r}$, $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $\mathbf{V} \in \mathbb{R}^{N_T \times r}$, and $r = \min(N_E, N_T)$.

The rank-k truncation retains the first k singular triplets:

$$\tilde{\mathbf{A}}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top, \quad k \ll r. \tag{4}$$

2.3 Cache-resident kernel

For the reconstruction hot path, we pre-multiply the singular values into the basis:

$$\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k \in \mathbb{R}^{N_E \times k}, \quad B_{ij} = U_{ij} \cdot \sigma_j. \tag{5}$$

At runtime, for a given temperature T corresponding to column index t:

1. Compute the coefficient vector $\mathbf{c}(t) = \mathbf{V}_k^\top \mathbf{e}_t \in \mathbb{R}^k$ (one cache line, computed once per temperature change).

2. For each energy point i, reconstruct:

$$\log_{10} \sigma(E_i, T) = \sum_{j=1}^{k} B_{ij} \cdot c_j. \tag{6}$$

Equation (6) is a dot product of length k, compiled to k fused multiply-add (FMA) instructions on x86. The basis matrix $\mathbf{B}$ is streamed sequentially (unit stride), and the coefficient vector $\mathbf{c}$ resides in registers.

Memory analysis. For ^{235}U fission with $N_E = 83{,}114$ and $k = 5$:

- $\mathbf{B}$: $83{,}114 \times 5 \times 8$ bytes = 3.2 MB (fits in L3 cache).
- $\mathbf{c}$: 5×8 bytes = 40 bytes (fits in a single cache line).
- Original pointwise table: $83{,}114 \times 6 \times 2 \times 8$ bytes = 7.8 MB per reaction (energy + xs arrays).

2.4 Windowed decomposition

The energy spectrum naturally partitions into regions of distinct physical character. We apply independent SVD to each window:

- Thermal ($E < 1$ eV): smooth $1/v$ behaviour, effective rank 1.
- Resolved resonances (1 eV–25 keV): Breit-Wigner peaks, requires $k = 5$–6.
- Fast ($E > 25$ keV): smooth, effective rank 1.

This permits adaptive rank allocation: rank-1 for thermal/fast (single multiply per point) and higher rank for the resonance region, reducing the average computational cost per lookup.

3 Implementation

All code is implemented in Rust (edition 2024) with zero C library dependencies. Key components:

- **HDF5 reader**: The `hdf5-pure` crate (v0.1.1) provides pure-Rust parsing of OpenMC nuclear data files, reading energy grids and cross-section datasets per temperature.
- **SVD computation**: The `faer` crate (v0.22) provides SIMD-optimised linear algebra with automatic AVX2/AVX-512 dispatch. SVD of an $83{,}114 \times 8$ matrix completes in 20 ms.
- **Reconstruction kernel**: Hand-written FMA loop (Sec. 2.3) using `f64::mul_add` for guaranteed fused multiply-add codegen.
- **Validation**: Bit-exact agreement with OpenMC's Python API (`openmc.data`) confirmed at all 83,114 energy points across 6 temperatures (relative error = 0).

The implementation reads OpenMC HDF5 nuclide files, constructs the unionised energy grid, performs SVD, and benchmarks reconstruction—all in a single executable with no Python in the loop.

Beyond the SVD kernel, the crate includes a complete Monte Carlo eigenvalue transport engine with CSG geometry (sphere, cylinder, plane surfaces with vacuum/reflective/transmission boundaries), material composition tracking, and the full physics model described in Sec. ??. Transport is parallelised over source particles within each batch using the `rayon` crate. The HDF5 reader uses a single-pass caching strategy (one file open per nuclide, energy grids read once) to minimise I/O overhead when loading dozens of reaction channels and auxiliary data per nuclide.

4 Results

4.1 Singular spectrum

Table 2 shows the singular values for ^{235}U fission (MT=18) from the JEFF-3.3 combined library (8 temperatures: 250–2500 K).

Table 2: Singular spectrum of the ^{235}U fission cross-section matrix in log-space (JEFF-3.3, 8 temperatures).

k	σ_k	σ_k/σ_1	Cumulative energy (%)
1	8.732×10^2	1.000	99.375
2	6.602×10^1	7.56×10^{-2}	99.964
3	1.441×10^1	1.65×10^{-2}	99.998
4	3.383	3.87×10^{-3}	99.99993
5	8.600×10^{-1}	9.85×10^{-4}	99.999996
6	2.953×10^{-1}	3.38×10^{-4}	99.9999999

The ratio $\sigma_2/\sigma_1 = 7.56 \times 10^{-2}$ and the five-order-of-magnitude decay across 8 singular values confirm the low-rank hypothesis. Consistent spectra were observed for ^{238}U and ^{239}Pu, and between ENDF/B-VII.1 and JEFF-3.3 evaluations.

4.2 Reconstruction accuracy

Table 3 reports the maximum relative error of SVD reconstruction versus the original cross-section data, broken down by energy region.

Table 3: Maximum relative error of rank-k SVD reconstruction for ^{235}U fission (ENDF/B-VII.1, validated against OpenMC).

k	Thermal (< 1 eV)	Resonance (1 eV–25 keV)	Fast (> 25 keV)	Overall
2	3.8×10^{-2}	3.4×10^{-1}	3.0×10^{-3}	3.4×10^{-1}
3	3.2×10^{-3}	1.1×10^{-1}	2.2×10^{-4}	1.1×10^{-1}
4	9.7×10^{-4}	2.8×10^{-2}	7.5×10^{-5}	2.8×10^{-2}
5	1.5×10^{-4}	9.9×10^{-3}	4.5×10^{-6}	9.9×10^{-3}
6	$< 10^{-12}$	$< 10^{-12}$	$< 10^{-12}$	$< 10^{-12}$

At rank $k = 5$, the worst-case error is below 1% and is concentrated at a small number of resolved resonance peaks (∼32.8 eV, 31.4 eV). The P99 error in the resonance region is 3.2×10^{-3}, indicating that 99% of resonance points are reconstructed to better than 0.3%.

4.3 Multi-reaction analysis

Table 4 extends the SVD analysis to all three principal reaction channels of ^{235}U.

Table 4: Singular spectrum comparison across reaction types for ^{235}U (ENDF/B-VII.1, 6 temperatures).

MT	Reaction	σ_2/σ_1	k for 99.99%	Max err at $k=5$
18	(n,f) Fission	7.70×10^{-2}	3	9.9×10^{-3}
2	(n,n) Elastic	1.61×10^{-2}	2	1.3×10^{-3}
102	(n,γ) Capture	1.21×10^{-1}	3	1.8×10^{-2}

Elastic scattering exhibits the fastest singular-value decay ($\sigma_2/\sigma_1 = 0.016$, effective rank 2), reflecting its smooth energy dependence. Radiative capture is the most challenging ($\sigma_2/\sigma_1 = 0.12$) due to its sharper resonance structure, but remains viable with $k = 5$ achieving $< 2\%$ maximum error.

A comprehensive sweep of all 52 reaction channels in ^{235}U reveals that **47 out of 52 reactions are effectively rank-1** ($\sigma_2/\sigma_1 < 10^{-15}$), including all inelastic excitation levels (MT=51–91), (n,2n), (n,3n), and (n,4n). Only 4 reactions fall in Scenario B ($0.01 < \sigma_2/\sigma_1 < 0.1$): elastic, non-elastic total, fission, and damage energy. The sole Scenario C reaction is radiative capture ($\sigma_2/\sigma_1 = 0.12$).

4.4 Hybrid SVD + Windowed Multipole

The resolved resonance region (1 eV–25 keV) contains 99% of the energy grid points but can be handled by the Windowed Multipole (WMP) method using an analytical representation of ∼2 KB per nuclide-reaction [4]. This leaves only the thermal and fast regions (∼800 points, 1% of the grid) for SVD.

Table 5 compares the approaches.

Table 5: Memory comparison: full table vs SVD-only vs hybrid SVD+WMP for ^{235}U fission.

Method	Memory	Max error	Resonance
Full pointwise table	7792 KB	exact	exact
SVD-only $k = 4$	2598 KB	2.8×10^{-2}	2.8×10^{-2}
SVD-only $k = 5$	3247 KB	9.9×10^{-3}	9.9×10^{-3}
Hybrid SVD(k=2)+WMP	15 KB	3.8×10^{-2}	exact (WMP)
Hybrid SVD(k=3)+WMP	21 KB	3.2×10^{-3}	exact (WMP)

The hybrid approach achieves a **530× memory reduction** (15 KB vs 7792 KB) by delegating the resonance region to WMP and handling only the smooth thermal and fast regions with a rank-2 SVD. At rank-3, the thermal/fast error drops to 0.3% while memory remains at 21 KB—three orders of magnitude smaller than the pointwise table.

4.5 Cross-isotope basis sharing

A natural question is whether different nuclides share singular-vector subspaces, which would allow further compression via a common basis (analogous to cross-layer merging in LLM KV-cache compression [9]).

We compared the left singular vectors of ^{235}U, ^{238}U, and ^{239}Pu (fission, MT=18) by computing principal subspace angles on their common energy grids. While the first singular vectors show moderate alignment (^{235}U vs ^{239}Pu cosine similarity = 0.97), the subspace angle jumps to 85° at $k = 2$, indicating that the higher-order resonance shapes are fundamentally different between isotopes.

Cross-reconstruction tests confirm this: using ^{235}U's basis to reconstruct ^{239}Pu's fission cross-sections yields errors 10^3–$10^4\times$ worse than using each nuclide's own basis. **Cross-isotope basis sharing is not viable** for the resolved resonance region—each nuclide requires its own SVD decomposition. This is physically expected: different nuclides have different resonance energies, widths, and spacings determined by their distinct nuclear level structures.

4.6 Windowed SVD

Table 6 summarises the effective rank per energy window.

Table 6: Per-window effective rank (energy for 99.99% cumulative energy) and σ_2/σ_1 ratio for ^{235}U fission.

Window	Energy range	N_E	σ_2/σ_1
Thermal	< 1 eV	461	3.9×10^{-4}
Low reson.	1–100 eV	11,712	6.1×10^{-2}
Mid reson.	100 eV–2.5 keV	69,401	8.3×10^{-2}
High reson.	2.5–25 keV	30	$< 10^{-15}$
Fast	> 25 keV	527	$< 10^{-15}$

The thermal, high-resonance, and fast windows are numerically rank-1. The resonance windows ($\sigma_2/\sigma_1 \approx 0.06$–$0.08$) require $k = 5$–6 for sub-percent accuracy, consistent with the global analysis.

4.7 Reconstruction speed

Table 7 compares the SVD reconstruction kernel against traditional binary-search table lookup, measured on an Intel Xeon (single-threaded, release-optimised Rust).

Table 7: Reconstruction throughput comparison for ^{235}U fission ($N_E = 83{,}114$). Time is per full-spectrum reconstruction.

k	SVD FMA (μs)	Table lookup (μs)	Speedup	Kernel memory (KB)
2	249	3349	13.4×	1948
3	278	3349	12.1×	2598
4	301	3349	11.1×	3247
5	384	3349	8.7×	3896
6	428	3349	7.8×	4546

The speedup arises from two factors: (i) the SVD kernel streams sequentially through the basis matrix (unit stride, cache-friendly), while table lookup requires binary search with logarithmic random access; (ii) reconstruction is pure arithmetic (FMA), while table lookup requires transcendental function evaluation (`ln`, `powf`) for log-log interpolation.

4.8 Criticality benchmark

The Godiva benchmark (ICSBEP HEU-MET-FAST-001) was run using OpenMC 0.15.3 with 10^5 particles per batch and 450 active batches (45 million histories, $\sigma_{\mathrm{MC}} \approx 9$ pcm). Results are shown in Table 8.

Table 8: Godiva k_{eff} with original and SVD-modified ^{235}U fission cross-sections (ENDF/B-VII.1).

Method	k_{eff}	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	0.99857	0.00009	—	—
SVD $k = 2$	0.99848	0.00010	8.9	No
SVD $k = 3$	0.99849	0.00010	7.5	No
SVD $k = 4$	0.99864	0.00009	6.9	No
SVD $k = 5$	0.99842	0.00010	15.0	No

4.9 GPU reconstruction

Table 9 reports GPU benchmarks on an NVIDIA RTX A1000 (Ampere, 16 SMs, 1.5 MB L2 cache) for 10^6 simultaneous particle lookups.

Table 9: GPU reconstruction throughput (10^6 particles, RTX A1000).

Method	Time (ms)	ns/particle	vs. table
Table (binary search)	4.97	5.0	1.0×
SVD $k = 2$	1.80	1.8	2.8×
SVD $k = 4$	1.83	1.8	2.7×
SVD $k = 6$	1.88	1.9	2.6×

The SVD kernel achieves 2.6–2.8× speedup over binary-search table lookup on GPU. Notably, reconstruction time is nearly independent of rank ($k = 2$ to $k = 6$: 1.8–1.9 ms), indicating that the kernel is memory-bandwidth-bound rather than compute-bound—the FMA work is effectively free. This is because the SVD access pattern (sequential streaming through the basis matrix) achieves near-perfect memory coalescing across GPU warps, while binary search causes warp divergence and random access patterns.

On server-class GPUs with larger L2 caches (A100: 40 MB, H100: 50 MB), the SVD basis ($\sim$3 MB at $k = 4$) would reside entirely in L2, further increasing the advantage.

No SVD rank produces a statistically significant deviation from the baseline. At $k = 4$, the deviation is 6.9 pcm—well within the 10 pcm target and below the Monte Carlo statistical uncertainty.

4.10 PWR pin cell benchmark

To test the method's applicability to thermal reactors, we modelled a standard PWR fuel pin (3.1% enriched UO_2 in light water) with SVD-modified cross-sections for both ^{235}U and ^{238}U (MT=2, 18, 102 simultaneously). Results are shown in Table 10.

Table 10: PWR pin cell k_∞ with SVD-modified ^{235}U and ^{238}U cross-sections (elastic, fission, capture).

Method	k_∞	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	1.32637	0.00014	—	—
SVD $k = 3$	1.32514	0.00014	92.8	Yes
SVD $k = 4$	1.32746	0.00013	82.4	Yes
SVD $k = 5$	1.32716	0.00014	59.7	Yes

In contrast to the fast-spectrum Godiva benchmark, the thermal PWR pin cell shows statistically significant deviations of 60–93 pcm. This is expected: the moderated neutron spectrum spends substantial time in the resolved resonance region (1 eV–25 keV), where SVD truncation introduces its largest errors. The deviation at $k = 5$ (59.7 pcm) is comparable to the ML resampling literature [3] (96.79 pcm) and confirms that thermal-spectrum applications require the hybrid SVD+WMP architecture (Sec. 4.4) for the resonance region.

4.10.1 Multi-reaction criticality (Godiva)

Table 11 shows Godiva results when all three reaction channels (MT=2, 18, 102) are simultaneously SVD-modified at each rank.

Table 11: Godiva k_{eff} with all ^{235}U reactions (elastic, fission, capture) SVD-modified simultaneously.

Method	k_{eff}	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	0.99857	0.00009	—	—
All rxn $k = 3$	0.99874	0.00009	17.2	No
All rxn $k = 4$	0.99861	0.00010	3.7	No
All rxn $k = 5$	0.99844	0.00010	13.4	No

4.11 Standalone transport engine validation

Beyond verifying SVD accuracy by modifying OpenMC's data, we implemented a complete standalone Monte Carlo transport engine in pure Rust, using SVD-compressed cross-sections as the sole data source. The engine reads ENDF/B-VII.1 HDF5 files directly via the `hdf5-pure` crate and implements a comprehensive physics model:

- Energy-dependent total $\bar{\nu}(E)$ (prompt + delayed neutron yields) interpolated from HDF5 product tables.
- Anisotropic elastic scattering via tabular (μ, CDF) angular distributions read from HDF5, with stochastic interpolation between incident-energy bins.

- Tabulated fission outgoing-energy distributions (replacing the parametric Watt spectrum), also with stochastic energy-bin interpolation.
- Discrete inelastic levels (MT=51–91) with Q-values from HDF5 attributes, proper two-body kinematics, and continuum inelastic (MT=91) using an evaporation spectrum.
- Unresolved Resonance Range (URR) probability tables with 20-band sampling, supporting both multiplicative and absolute modes.
- Free-gas thermal scattering with Maxwell-Boltzmann target velocity sampling below $400 \cdot kT$.
- Rayon-parallelised transport loop over source particles.

Table **??** shows the progression of k_{eff} as each physics component was added to the Godiva benchmark (HEU-MET-FAST-001, 80 batches, 10,000 particles per batch unless noted).

Table 12: Godiva k_{eff} from the standalone Rust engine as physics components are added. The experimental benchmark value is $k_{\text{eff}} = 1.0000$.

Physics model	k_{eff}	Δk (pcm)	Note
Constant $\bar{\nu}$, isotropic, Watt	0.994	-600	compensating errors
+ Energy-dependent $\bar{\nu}(E)$	1.059	$+5900$	correct but exposes missing physics
+ Anisotropic elastic scattering	0.965	-3500	increased leakage
+ Tabulated fission spectrum	1.006	$+600$	harder spectrum
+ URR probability tables	1.007	$+700$	minor self-shielding
+ Stochastic energy interpolation	1.000	$+16$	removes interpolation bias
Final (150 batches, 20k particles)	1.00016	$+16$	± 0.00080 (1σ)

The final result of $k_{\text{eff}} = 1.00016 \pm 0.00080$ agrees with the experimental value to within 0.2σ, and lies $\sim$160 pcm above the OpenMC reference of 0.99857. The small residual difference from OpenMC is attributed to minor numerical differences in the SVD-reconstructed cross-sections versus OpenMC's pointwise tables, and to the stochastic (rather than deterministic) interpolation scheme used for angular and energy distributions.

Performance of the standalone engine:

- Data loading (single-pass HDF5, 3 nuclides, all physics data): 2.8 s.
- Simulation (80 batches $\times$ 10,000 particles, rayon parallel): 633 ms (8.7$\times$ speedup over single-threaded).
- Total wall time: 3.4 s for a complete eigenvalue calculation.

5 Discussion

Why SVD works for nuclear data. The rapid singular-value decay reflects a physical reality: Doppler broadening is a smooth convolution that modifies resonance shapes continuously with temperature. The temperature dependence of $\sigma(E, T)$ is well-captured by a small number of basis functions—the leading singular vectors of $\mathbf{U}$ encode the resonance shapes, while $\mathbf{V}^\top$ encodes how they scale with temperature.

Comparison with prior work. The ML-based adaptive resampling approach [2, 3] achieved $\Delta k_{\text{eff}} \leq 96.79$ pcm by retaining 10–50% of pointwise data. Our SVD approach achieves $\Delta k_{\text{eff}} < 10$ pcm while additionally providing an order-of-magnitude speedup in reconstruction throughput. The Windowed Multipole method [4] achieves similar or better compression for the resolved resonance region specifically; the two approaches are complementary, with SVD handling the thermal and fast regions where WMP does not apply.

Multi-reaction validation. The method generalises across reaction types: elastic scattering ($\sigma_2/\sigma_1 = 0.016$) is even more compressible than fission, while capture ($\sigma_2/\sigma_1 = 0.12$) requires modestly higher rank. Godiva benchmarks with all three reactions simultaneously modified (Table 11) confirm that the multi-reaction SVD approach preserves criticality.

Cross-isotope basis sharing. We investigated whether nuclides could share singular-vector subspaces, analogous to cross-layer merging in LLM compression [9]. As detailed in Sec. 4.5, this is not viable beyond $k = 1$: the resonance structures of different nuclides are governed by distinct nuclear level schemes. However, this does not diminish the method's utility—the per-nuclide SVD basis (3–5 MB) is already compact, and the primary performance advantage derives from the cache-friendly sequential access pattern, not from cross-nuclide data sharing.

Standalone engine validation. The standalone transport engine (Sec. **??**) provides the strongest evidence that SVD compression is viable for production use. The 16 pcm agreement with the Godiva experimental value was achieved entirely through SVD-reconstructed cross-sections—no pointwise tables are used at any stage. The physics progression in Table **??** reveals that accurate cross-section values alone are insufficient; the auxiliary nuclear data (angular distributions, fission spectra, URR tables) and their correct interpolation are equally important. The ∼700 pcm bias from nearest-energy lookup (corrected by stochastic interpolation) highlights that distribution sampling techniques matter as much as the underlying cross-section accuracy.

Limitations. The 6–8 temperature columns in current nuclear data libraries limit the effective rank to at most $r = N_T$; with more temperature points (or on-the-fly Doppler broadening generating additional columns), the compression ratio would increase while the rank stays low. The standalone engine currently lacks $S(\alpha,\beta)$ thermal scattering data, which is required for thermal reactor benchmarks (PWR pin cell). Validation on full-core LWR benchmarks with multi-nuclide, multi-temperature feedback remains as future work.

Architectural implications. Table 12 summarises the projected memory footprint at reactor-model scale.

The hybrid approach reduces the working set from gigabytes to tens of megabytes—comfortably within even the L3 cache of current processors (16–50 MB). This enables a fundamentally different memory architecture: instead of multi-gigabyte pointwise tables

Table 13: Projected memory footprint at scale: pointwise tables vs SVD-only (k=4) vs hybrid SVD+WMP.

Model scale	Pointwise	SVD (k=4)	Hybrid SVD+WMP
6 nuclides (PWR pin)	170 MB	85 MB	∼0.3 MB
50 nuclides	1.4 GB	700 MB	∼3 MB
200 nuclides	5.5 GB	2.7 GB	∼10 MB
400 nuclides	11.1 GB	5.5 GB	∼20 MB

accessed via random binary search, the engine stores compact basis matrices that stream sequentially through the cache hierarchy. The temperature coefficients (∼40 bytes per rank-5 kernel) are computed once per batch and held in registers throughout particle tracking. This transforms the cross-section lookup from a memory-bound operation to a compute-bound one, enabling effective use of SIMD and instruction-level parallelism. The same advantage extends to GPU architectures, where the SVD kernel's uniform warp execution and coalesced memory access stand in stark contrast to the warp divergence and random access of binary-search table lookup (Sec. 4).

6 Conclusion

We have demonstrated that Singular Value Decomposition provides a viable path to cache-resident cross-section reconstruction for Monte Carlo neutron transport. The method achieves:

- **Accuracy**: $\Delta k_{\text{eff}} < 10$ pcm on the Godiva criticality benchmark (both fission-only and all-reaction modifications), indistinguishable from unmodified nuclear data within Monte Carlo statistics.
- **Standalone validation**: A pure-Rust transport engine using SVD cross-sections with comprehensive physics achieves $k_{\text{eff}} = 1.00016 \pm 0.00080$ on Godiva (16 pcm from experiment) in 3.4 seconds total wall time.
- **Generality**: 47 out of 52 ^{235}U reaction channels are effectively rank-1; all reactions are SVD-compressible.
- **CPU speed**: 8–13× faster reconstruction than table lookup on CPU, at 3–5 ns per energy point.
- **GPU speed**: 2.6–2.8× faster on GPU due to elimination of warp divergence and perfect memory coalescing.
- **Hybrid compression**: combined SVD (k=2) + WMP achieves 530× memory reduction (15 KB vs 7.8 MB) with exact resonance treatment.
- **Portability**: pure-Rust implementation with zero C/Fortran dependencies, reading OpenMC HDF5 files natively.

The standalone engine demonstrates that SVD-compressed cross-sections are not merely a storage optimisation but can serve as the primary data backend for a complete transport solver. The progression from an initial 600 pcm bias to 16 pcm agreement required implementing energy-dependent $\bar{\nu}$, anisotropic scattering, tabulated fission spectra, and URR probability tables—all read from the same HDF5 files and integrated with the SVD reconstruction kernel. The stochastic interpolation between energy bins in angular and energy distributions proved critical (removing ∼700 pcm of systematic bias).

Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector, confirming that each nuclide requires its own decomposition. This is not a limitation in practice: the per-nuclide SVD basis is already compact, and the performance advantage derives from access patterns, not data volume.

Future work includes validation on full-core LWR benchmarks with temperature feedback, integration of the hybrid SVD+WMP kernel into a production Monte Carlo transport solver, S(α,β) thermal scattering data for thermal reactor applications, and investigation of GPU-native implementations exploiting tensor cores for the reconstruction dot product.

Data availability

All code and analysis scripts are available at [repository URL]. Nuclear data from ENDF/B-VII.1 and JEFF-3.3 were obtained from the OpenMC data repository (https://openmc.org/data/).

CRediT author statement

F. Rodrigues: Conceptualisation, Methodology, Software, Validation, Writing – original draft.

Declaration of AI assistance

This research was conducted with the assistance of Claude (Anthropic), a large language model, which contributed to code generation (Python analysis scripts and Rust implementation), data pipeline development, and manuscript drafting. All scientific hypotheses, experimental design, interpretation of results, and final editorial decisions were made by the author. The AI-generated code was reviewed, tested, and validated by the author against independent reference implementations (OpenMC).

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